



JAIDEV EDUCATION SOCIETY'S
J D COLLEGE OF ENGINEERING AND MANAGEMENT

KATOL ROAD, NAGPUR

Affiliated to DBATU, RTMNU & MSBTE, Mumbai.

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An Autonomous Institute, with NAAC "A" Grade

Department of Computer Science & Engineering

"A place to learn; A chance to grow"

Session: 2025-26



MSE-2 EXAMINATION

Course: B. Tech. in Computer Science & Engineering

Year/Semester: 5th Semester (3rd Year)

Subject: DBMS

Subject Code: CS5T001

Max Marks: 20

Date: 29/09/2025

Duration:- 1 Hr.

Instructions to the Students:

1. All the questions carry marks as indicate.
2. Assume suitable data whenever necessary.
3. Illustrate your answer whenever necessary with the help of neat sketches.

Students should be able to:

1. Learn and understand fundamentals of database management system.
2. Exhibit the query development knowledge.
3. Learn modelling and normalization of databases.
4. Learn query processing and optimization techniques.
5. Analyse the File Organization, Indexing and Hashing and exhibit the knowledge of transaction and concurrency control.

SECTION A

1. All Questions are compulsory.
2. Answer in one word/sentence.

Q. No.	Questions	Level/CO Mapped	Marks
Q.1	List out the Problems related to decompositions	L1/CO3	1
Q.2	State the term multi-valued dependencies.	L1/CO3	1
Q.3	Restate the principles of query optimization.	L1/CO4	1
Q.4	State the advantages of normalization.	L1/CO3	1

SECTION B

1. All Questions are compulsory.
2. Solve answer in brief (two, three Sentence)

Q.5	Express the term functional dependency? Explain its types in detail	L2/CO3	2
Q.6	Explain about Loss less-join & Lossy decomposition.	L2/CO3	2
Q.7	Compare 3NF and BCNF.	L3/CO3	2
Q.8	List out the query optimization strategies.	L1/CO4	2

SECTION C

1. Solve any two questions.

Q.9	Elaborate 1NF, 2NF, 3NF, 4NF and 5NF with Examples?	L4/CO3	4
Q.10	Interpret the steps involved in processing a query. Consider the Schema S = (V, W, X, Y, Z). Suppose the following F.D hold: Z-> V, W->Y, XY->Z, V->WX. State whether the following decomposition schema S is lossless join	L6/CO4	4
Q.11	decomposition or loss decomposition. i) S1 = (V, W, X) & S2 = (V, Y, Z) ii) S1 = (V, W, X) & S2 = (X, Y, Z)	L5/CO3	4